Total work at a specified solution error

Measured result: at absolute density L1 error ≤ 0.005, the sampled second-order method meets the cell-average target on 16^3 cells; the first-order method needs 64^3. It takes about **25.8× less complete-solve time** than the improved first-order implementation in this smooth test. This is an empirical comparison, not a general optimality theorem.

The exact problem

A periodic unit cube contains a gamma=2 ideal gas with pressure 1, velocity (0.5, 0.2, −0.1), and density $1 + 0.1 \sin[2\pi(x+y+z-0.6t)]$. All five conserved variables are evolved to T=0.05. This is a transported entropy wave embedded in 3D, not a general nonlinear-interaction or shock test. It is an exact solution of the mathematical PDE; no experimental validation of the physical model is implied.

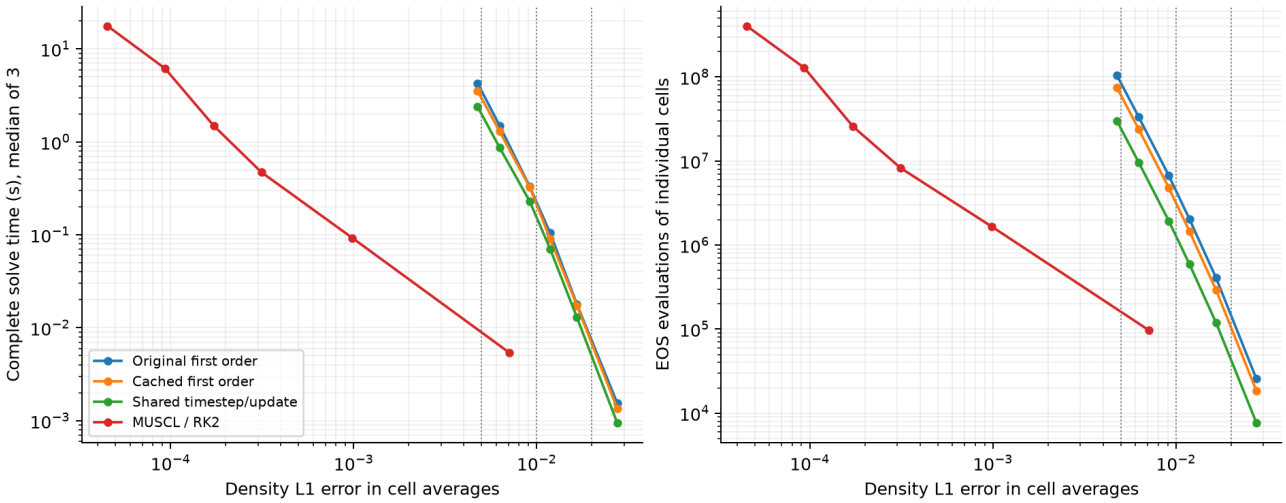
The primary ruler compares exact cell averages: the sinusoidal amplitude is multiplied by $\mathrm{sinc}(1/n)^3$. Error targets 0.02, 0.01 and 0.005 are absolute nondimensional density errors. Target 0.005 is 0.5% of the mean density, not 0.5% relative to the perturbation. A separate reconstructed-field audit checks the representation issue.

Method at target 0.005	Grid	Actual error	Median solve (s)
Original first order	64^3	0.004750	4.2736
Cached first order	64^3	0.004750	3.5287
Shared timestep/update	64^3	0.004750	2.3885
MUSCL / RK2	16^3	0.000983	0.0926

What changed from the earlier caching result

The full solve now counts initial-data construction, timestep selection, all stages, allocation, and per-step admissibility checks. The new first-order implementation shares thermodynamics between timestep selection and the update. The higher-order comparator uses MC-limited conservative reconstruction, Rusanov fluxes and SSPRK2. It intentionally changes the discretization; this is a standard-method comparison, not discovery of a novel algorithm.

The measured error–work frontier



Each curve contains all six tested grids. Dashed lines mark the three tolerances. Lower error lies to the left. Timing is a median of three trials with shuffled method order; all trials are retained. These are candidate implementations on one non-isolated CPU host, not best-in-class production solvers.

Target	First-order grid	MUSCL/RK2 grid	Status
0.02	16^3	8^3	Cell-average criterion
0.01	32^3	8^3	Cell-average criterion
0.005	64^3	16^3	Cell-average criterion

Accounting is part of the ruler

At target 0.005, the improved first-order solve uses 14,942,208 cell-stage updates and 30,146,560 EOS cell evaluations. MUSCL/RK2 uses 122,880 cell-stage updates and 1,662,976 EOS cell evaluations. A MUSCL stage costs more than a first-order stage: these proxies are not equal-cost FLOPs. Instrumentation independently verifies the EOS counts.

Including final analytic error analysis gives median solve-plus-analysis times of 2.4183 s and 0.0931 s, respectively. Program startup, file writing and orchestration are outside these times and visible in the lab receipts.

The primary 72-solve calibration sweep itself cost 123.1 s of solver time, 0.63 s of error analysis, and 123.9 s of lab-job wall time. It performed 315,095,040 cell-stage updates. Calibration and verification are not free and are not amortized into the selected single-solve times.

All three first-order variants have identical recorded solution and timestep hashes. The second-order results have different errors and step counts. General positivity is not proved: reconstructed states, RK stages and completed steps are checked and invalid runs would be rejected.

Physical representation changes admission

A small error in cell averages can hide subcell structure. The audit applies the same MC-limited piecewise-linear density reconstruction to selected outputs, then integrates absolute error numerically with tensor Gauss quadrature. Orders 3 and 5 are compared; this is not an interval-certified error bound.

Method / grid	Average L1	Reconstructed L1	Target / pass
Shared timestep/update / 16^3	0.016557	0.017122	0.02 / yes
Shared timestep/update / 32^3	0.009135	0.009175	0.01 / yes
Shared timestep/update / 64^3	0.004750	0.004757	0.005 / yes
MUSCL / RK2 / 8^3	0.007091	0.011755	0.01 / no
MUSCL / RK2 / 16^3	0.000983	0.001769	0.005 / yes

The 8^3 MUSCL result passes projected error 0.01 but fails reconstructed-field error 0.01. It must not be advertised as satisfying the latter. The selected 64^3 first-order and 16^3 second-order results still meet the numerical reconstructed target 0.005. The reconstruction cost is included in the separate single-solve audit; those single samples are not used as repeated timing evidence.

A sharper negative control has zero cell-average error on a one-cell grid but reconstructed density L1 error about 0.0637. This demonstrates why output representation, resolution and error norm must all be specified before using a performance claim.

Reference and implementation audits

An independent 8-point-per-axis quadrature checks the analytic cell-average formula to 2.22e-16 on the sampled cells. Comparing point samples with cell averages can differ by 0.018; the benchmark does not confuse the two. A run stopping before T=0.05 is rejected, and a coarse first-order run fails its tolerance as expected.

Measured pressure and velocity drift remain at roundoff scale, and conservation diagnostics are retained for every case. These checks support the smooth-wave experiment. They do not validate shock accuracy, arbitrary physical observables, or the Euler model against laboratory measurements.

What is proved—and what is still open

Twelve named theorems passed Lean and transitive-axiom audits in the bounded Simjecture executor. Only propext, Classical.choice and Quot.sound appear in the recorded dependencies. The theorem statements and hypotheses remain the authority.

Formal artifact	Checked result	Scope
ContactReference.lean	Pressure/velocity embedding, algebraic residual reduction and component-error factors	Density derivatives are supplied to the residual identity
WaveDerivative.lean	Affine-phase sine derivative and transport-phase cancellation	Scalar analytic reference ingredients
ErrorWork.lean	Convex-stencil nonexpansiveness; error accumulation; error/work conversion	Conditional hypotheses, not discharged for nonlinear Euler codes

The conditional rule is: if initial error is zero, each step contributes at most C/n^2, and the run takes at most $K\,n$ steps, then final error is at most $K\,C/n$. With $m \le k\,n$, the modeled work satisfies $n^3 m \le k\,n^4$. A desired epsilon can therefore define a sufficient grid when the constants and hypotheses have actually been certified.

Those stability and local-defect hypotheses have not been established for the implemented Rusanov or MUSCL/RK2 solvers. No measured convergence curve is promoted to a proof. The uniform-error lemma is also distinct from the measured L1 norm. There is no unconditional error certificate, floating-point refinement, or matching global lower bound here.

Replay and next proof target

```
.venv/bin/python demos/lean_euler/accuracy/run_local.py
--output artifacts/lean-euler/accuracy-replay-001
```

Use a new output directory after installing the Lean capability. The replay runs the primary timing sweep before audit/proof workloads and retains all receipts, hashes, errors and failures. The next proof task is a certified stability/defect bound for a restricted Euler family or a certified case-specific error estimate. A smooth nonlinear vortex and validated shock test are the next empirical extensions.

Sources: Clawpack's Euler/contact discussion; official discussion of accuracy versus per-step cost. We do not benchmark Clawpack here. Code, data and proof receipts: draft PR #90. This is scripted assistant-authored research without independent scientific review.